Policy-gradient experiment assumptions

A record of environment choices, training conditions, evaluation definitions, and limits for the PG comparisons.

Scope. This note documents the exploratory six-seed A2C critic comparison and earlier PG benchmarks. It does not claim convergence, a new algorithmic guarantee, or scientific novelty. The concise results preview remains a local copy.

Reward and reporting safeguards

- Game transition and reward rules were not changed to improve results. No reward shaping, win bonus, altered terminal reward, or fabricated run was used.
- Training entropy is fixed at 0.01 in the new comparison. It is an actor regularizer, not a soccer reward term.
- Win rate and Nash gap answer different questions. The exact best response optimizes expected discounted score difference, not the probability of winning. Win rate is an empirical outcome frequency.
- All completed seeds are retained. The new seeds were selected before their outcomes to extend an already inspected pilot, so they are exploratory and not a preregistered holdout.

Environment assumptions

Choice	Setting	Interpretation
Board	Deterministic A10; 7 by 5; goal rows 1-3	Simultaneous U/D/L/R actions; no random move order in the new comparison.
Reward	+1 for P0 scoring; -1 for conceding; 0 otherwise	First goal ends the episode. No dense reward or shaping is applied.
Horizon	100 steps; gamma = 0.9	Policies receive remaining horizon; the exact reference solves the matching finite-horizon game.
Other formats	First-goal random move order; fixed-horizon goal reset	Earlier results are shown separately. Random order means stochastic move order, not a sampled board layout. Goal-reset outcomes are goal difference over the horizon.
Seats	P0 and P1 reported separately	The 50:50 column is an unweighted average of the two seat-specific win frequencies.

Training and evaluation choices

How the comparison was run

Methods and training budget

- The six-seed A10 comparison uses seeds 4-9; 2,000 iterations; 64 episodes per collected batch; five full-batch update passes; fictitious play with a policy snapshot every five iterations; entropy coefficient 0.01; and 1,000 evaluation games per seat and seed.

- **Standard A2C** is context and uses the repository's 10-step bootstrapped advantage with a learned online critic.

- **A2C MC learned-critic control** updates the actor with sampled full discounted returns minus a state-value baseline. Its value network learns from those sampled returns.

- **A2C frozen exact-value baseline** fits a state-time value network to exact solver values for 10 epochs, freezes it, and updates the actor with sampled full returns minus that baseline. The actor does not query exact action values. The separate oracle-advantage method is not this experiment.

- In fictitious play, each player trains against an opponent snapshot sampled uniformly from its past; the reported policy averages action probabilities across that player's snapshots. Networks are independent, and mirror symmetry is not imposed.

Evaluation definitions

- **Nash gap / exploitability** is the sum of both players' gains in expected discounted score when each replaces its policy with an exact best response to the other. It is measured in payoff units; zero means equilibrium for this finite-horizon evaluator.

- **Win rate vs. best response** counts wins in sampled games against the exact score-best response. It is not the objective minimized by that response and is not a substitute for Nash gap.

- The reported rates average 1,000 simulated games per player seat and seed. Game outcomes include tie and loss; win bars do not encode exploitability.

- Seed labels are matched across methods, but trajectories are not identical. Exact-value pretraining consumes random draws before actor updates, and the learned policies can visit different states.

Observed results and limits

What the evidence supports

Method	Seeds	Mean gap (range)	P0 wins vs. BR	P1 wins vs. BR
Standard A2C, 10-step	4-9; n=6	0.869 (0.839-0.898)	1.1%	0.8%
A2C MC learned baseline	4-9; n=6	0.854 (0.748-0.940)	1.5%	0.8%
A2C frozen exact-value baseline	4-9; n=6	0.874 (0.808-0.928)	0.8%	0.8%

Interpretation. The frozen-value mean gap is 0.020 higher than the learned-MC control, but the seed-level direction is split 3 to 3. With six exploratory seeds and near-zero best-response win rates, there is no reliable evidence that exact-value pretraining improves equilibrium quality. All learned policies remain far from gap 0.

Run audit

Seeds 7-9 used the same protocol as seeds 4-6 and were selected before their outcomes. Two earlier local attempts were interrupted after temporary evaluator-harness errors; those attempts produced no saved result and are excluded. The final vectorized deterministic evaluator was checked against the project evaluator for both players and outcome metrics on a short horizon. Its cached critic construction matched the project exact critic array-for-array on a small game. The reported full-horizon rows come from the completed final runner.

Literature context

Prajapat et al., *Competitive Policy Optimization*, UAI 2021, proposes game-aware updates for competitive Markov games and includes a soccer study. Its setup differs from this project. Giannou et al., NeurIPS 2022, prove policy-gradient convergence under stated conditions; those conditions have not been established for these neural runs.

Raw new runs: `experiments/pg_paired_new_seeds_a10.csv`. The concise PG preview is intentionally kept local pending review. No publication or generalization claim follows from this small pilot.

Assumptions record · soccer transition and reward configuration held fixed · all values are exploratory.